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Summary Sheet

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From Minutes on the Court to Cash in the Bank An Inner-Outer Loop Decision-Making Framework

Summary

Amidst the deepening commercialization of professional sports, team management faces the dual challenge of achieving competitive success and financial profitability. Taking the Indiana Fever of the WNBA as a case study, this paper constructs an **Inner-Outer Loop Framework** that deeply integrates competitive performance with economic outcomes, aiming to maximize team profits and long-term value.

For the core decision system (Task 1), a **dual-layer dynamic model** is established. The **outer-loop model** serves as the decision hub, using discounted long-term valuation as the primary metric while comprehensively considering **(cash flow)**, debt service coverage ratio **(DSCR)**, and **Conditional Value at Risk(CVaR))**. The **inner layer** employs **Heuristic ADP** (Approximate Dynamic Programming) and **Monte Carlo simulation** to map specific tactical patterns and minute allocation to win probability, game entertainment value, and player health status. Data flows between layers enable closed-loop management spanning from roster rotation to top-level financial risk.

For player valuation and recruitment (Task 2), an innovative **dual-track valuation system** was established, using **entropy weighting** to quantify both athletic and commercial value. Through **synergy and redundancy matrices**, the analysis identifies roster weaknesses. Results indicate Indiana Fever's current backcourt lineup, driven by core players, has reached "saturation" with diminishing marginal returns. Therefore, priority is given to stabilizing the frontcourt through draft picks, free agency, and trade decisions to optimize roster balance.

Addressing league expansion impacts (Task 3), this study applies **spatial economics' gravity model** to quantify new teams' revenue dilution and erosion effects on existing markets, driving endogenous drift in peripheral weights. Sensitivity analysis reveals Chicago—with its massive economic scale and geographic proximity—poses the greatest threat as a potential expansion site. Consequently, the model proposes a **"Moat Strategy"**: when market value is compromised, defend brand equity by recruiting commercial superstars and sacrificing short-term profits.

For ticketing operations and contingency management (Tasks 4 and 5), this paper establishes a **dynamic ticket pricing model** based on game excitement levels. By simulating demand elasticity across different tactical phases, it determines optimal pricing paths for each game. Simultaneously, when key players face injury impacts, the model reoptimizes minute allocation for the remaining schedule via a risk engine. Utilizing a **Load- Balancing strategy**, it controls tail risks while ensuring operational robustness.

Research conclusions indicate this model reveals **the Pareto Frontier** of risk and reward. Higher economic returns often accompany greater health risks, while scientific load management effectively smooths the revenue curve.

Keywords: Inner-Outer Layer Coordination; Dual-Track Value Assessment; Gravity Model; Heuristic ADP; CVaR

Contents

1. Introduction **2**
1.1. Problem Background 2
1.2. Restatement of the Problem 2
1.3. Our Work 2

2. Model Preparations **3**
2.1. Assumptions and Justifications 3
2.2. Notations 4

3. Model **4**
3.1. Model Overview: From Minutes on the Court to Cash in the Bank 4
3.2. Value Model 5
3.3. Market Model 8
3.4. Inner-Layer Game Model 12
3.5. Financial Module 17
3.6. Potential for Future Development ΔFV_t 21
3.7. Outer Layer Overall Objective 22

4. Sensitivity Analysis and Robustness Testing **22**

5. Model Evaluation **23**
5.1. Advantages 23
5.2. Disadvantages: 23

1. Introduction

1.1. Problem Background

In professional sports, traditional evaluations emphasize a team's competitive success while overlooking the decisive constraints imposed by financial profitability in the era of sports economic commercialization. Taking the WNBA as an example, the league is currently undergoing a series of financial transformations. To navigate these changes, teams require systematic quantitative modeling and optimization methods to achieve an explainable trade-off between competitive performance and commercial profit.





Figure 1: WNBA Photo Highlights

1.2. Restatement of the Problem

1.2.1. *Problem 1: Establish a dynamic decision-making model spanning the entire season, enabling management to maximize team profits and long-term value while balancing risks and cash flow during roster adjustments and performance optimization.*

1.2.2. *Problem 2: Develop a practical player valuation methodology to assess the advantages, costs, and potential risks of this recruitment strategy for team operations.*

1.2.3. *Problem 3: Quantify the revenue dilution, intensified competition, and market redistribution resulting from league expansion. Recommend how teams should adjust initial strategies accordingly. Simultaneously analyze the impact of new team locations on the home team's revenue and decision-making, examining both favorable and unfavorable scenarios.*

1.2.4. *Problem 4: Develop a dynamic ticket pricing model for home games to determine optimal pricing strategies on a game-by-game (or phased) basis throughout the season.*

1.2.5. *Problem 5:When key players suffer injuries (or experience health declines), incorporate this as a performance shock into the model. Reoptimize subsequent game schedules, player rotation, and operational strategies to maximize profits and value while controlling injury risks and tail risks.*

1.3. Our Work

To avoid complicated description and intuitively reflect our work process, the flowhart is shown as the following Figure 2.

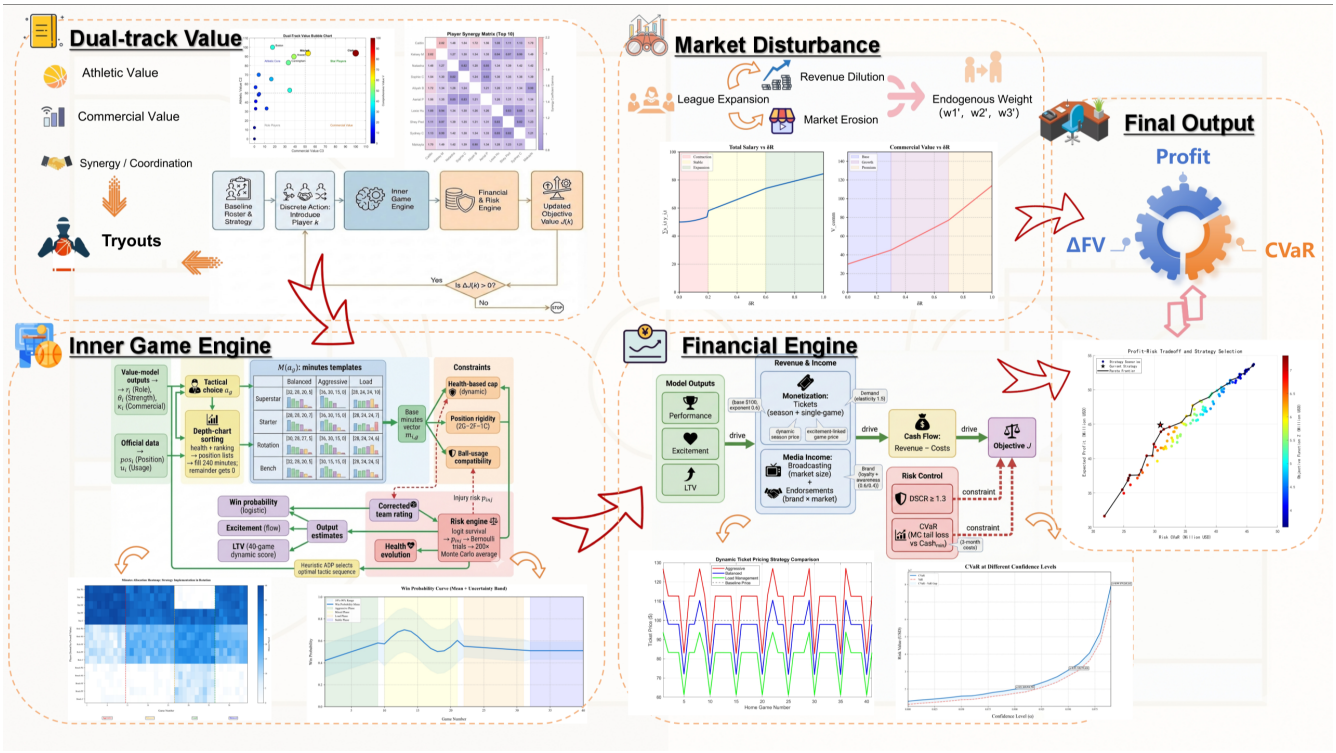


Figure 2: Flowchart

2. Model Preparations

2.1. Assumptions and Justifications

Before constructing our model, we establish the following assumptions based on practical considerations:

Assumption 1: Schedule and Opponent Strength are Exogenous

Justification: The schedule is set by the league, and opponent strength can be derived from publicly available data, remaining unaffected by individual team strategies in the short term.

Assumption 2: Dual-Track Value is Quantifiable

Justification: Both competitive and commercial value can be measured using public/proxy metrics, which can be standardized for comparison and weighted synthesis.

Assumption 3: Minutes and Position Constraints Hold

Justification: Total minutes per game are fixed at 240, and the basic position structure represents a hard constraint of realistic rotations.

Assumption 4: Injuries are random and vary with workload

Justification: Injuries are unpredictable but correlate with workload and health status, making

probabilistic modeling appropriate.

Assumption 5: Revenue mapping is used for relative comparison

Justification: Revenue is influenced by multiple factors, but win rate/excitement provides stable demand directionality, making it suitable for scenario comparison.

2.2. Notations

Table 1: Notations

Symbol	Meaning	Unit/Value
$i \in N$	Player Index	—
$g = 1, \ldots, 40$	Match Index	—
$a_g \in A$	Game g Tactical Mode	Category
$m_{i,g}$	Player i Playing Time in Game g	min
V_i^{perf}	Player i Athletic Value	—
V_i^{comm}	Player i Commercial Value	—
V_i^{total}	Player i Overall Value	—
WinProb_g	Game g Win Probability Prediction	$[0, 1]$
Excite_g	Game g Spectator Appeal Metric	—
R_g^{ticket}	Game g Ticket Revenue	Currency/Index
$H_{i,g}$	Player i Health Status in Game g	$[0, 1]$
$p_{i,g}^{\text{inj}}$	Player i Injury Probability in Game g	$[0, 1]$
Q_g	Opponent Strength	—
S_g	Schedule Variables	—
α	CVaR Confidence Level	$(0, 1)$

3. Model

3.1. Model Overview: From Minutes on the Court to Cash in the Bank

This paper constructs a model titled "An Inner–Outer Loop Decision Framework for Professional Sports Team Management" [5]deeply integrating athletic performance with economic outcomes. The outer layer of the model handles roster decisions, while the inner layer allocates minutes and outputs performance metrics, exposure, and injury pathways. These outputs are then fed into the outer layer to calculate profit, cash flow, DSCR, CVaR, and valuation, forming an actionable management system.

$$\max \mathbb{E}\left[\sum_{t=0}^{T} \frac{1}{(1+r)^t}(w_1\Pi_t + w_2\Delta FV_t - w_3 CVaR_\alpha(L_t))\right]$$

The outer-layer overall objective $r = r_f$ represents the discount rate, w_1, w_2, w_3 calculated as the

average long-term Treasury bond yield. It undergoes dynamic adjustments based on the **market model**, with different expansion policies and new team locations exerting varying influences on it, thereby impacting decision-making. Π_t, ΔFV_t *and* $CVaR_\alpha(L_t)$[1] are incorporated into subsequent financial models and serves as the core indicator of this model.

Considering the star factor and data availability, this paper selects the Indiana Fever team from the WNBA for analysis.

3.2. Value Model

To formulate next season's player acquisition strategy based on team needs, this paper establishes a dual-track value + calibration + synergy/redundancy value system standard.

3.2.1. *Data Processing: Metric Standardization*

This article will convert PER, WS, BPM, and other statistical metrics $Stat_i^{(k)}$ from the official website into z-scores:

$$\widetilde{Stat}_i^{(k)} = \frac{Stat_i^{(k)} - \mu_k}{\sigma_k}$$

μ_k, σ_k represent the mean and standard deviation of metric k across all players in the league during the same season (calculated directly from raw data). The z-score resolves issues arising from different measurement units across metrics.

3.2.2. *Dual-Track Value[5]*

This paper establishes two distinct value systems for evaluating players, enabling us to achieve dual success in both win rate and commercial performance.

$$V_{\text{perf}} = \sum_{k \in K} \omega_k \cdot \text{Stat}_i^{(k)} \qquad V_{\text{comm}} = u_1 \cdot (1 + F_i) + u_2 \cdot T_i + u_3 \cdot A_i$$

Competitive value V_{perf} is quantified by a set of performance metrics K, including PTS (Points), TRB (Total Rebounds), AST (Assists), and FG3% (3-Point Field Goal Percentage), among others, to measure player performance.ω_k represents the weighting assigned to each metric.

Commercial value V_{comm} is derived from **social media engagement** F_i, estimated by fitting Instagram follower counts and Wikipedia traffic. Google Trends indices represent **search popularity** T_i. Indicators such as whether jersey merchandise enters Top Sellers or secures national-level brand endorsements reflect personal **brand value** A_i. u_1, u_2, u_3 reflect the relative importance of different commercial factors.

u_1, u_2, u_3 *and* ω_k calculate weights using the entropy weight method[2] based on the **discreteness of each indicator**, ensuring that indicators with high discriminative power are assigned greater weight and avoiding bias from manually setting weights.

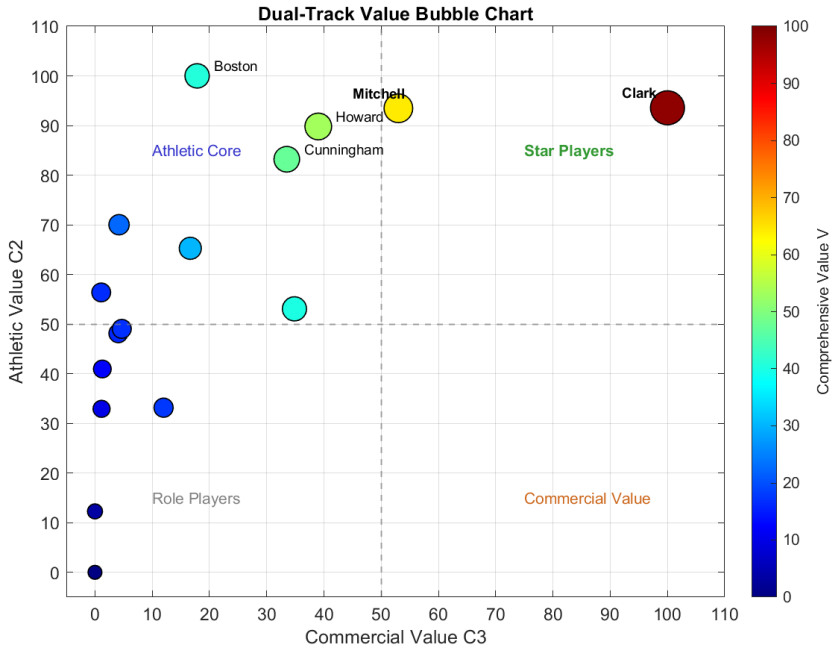


Figure 3: Value-based processing results

The chart above shows the results of value-based processing, revealing the value ranking of all players. This highlights Clark's pivotal role as the undisputed star player within the team.

3.2.3. *Comprehensive Value*

Comprehensive value is calculated by weighting competitive value and commercial value.

$$V_{\text{total}} = \alpha \cdot V_{\text{perf}} + (1 - \alpha) \cdot V_{\text{comm}}$$

α represents the weight of competitive value and commercial value within the composite value. This paper employs the coefficient of variation method for weight calibration, where a higher weight corresponds to greater discriminative power of the indicator, reflecting the emphasis placed on individual value within the team.

3.2.4. *Synergy and Redundancy*

To prevent the pseudo-optimality of "stacking homogeneous players," this paper introduces $Syn(\mathbf{y})$ quadratic penalty term $Red(\mathbf{y})$ for synergy and positional redundancy:

$$Syn(\mathbf{y}) = \sum_{i<j} \Gamma_{ij} y_i y_j \qquad Red(\mathbf{y}) = \sum_{p} \kappa_p \cdot \max\left(0, \sum_{i:pos(i)=p} y_i - U_p\right)^2$$

Γ_{ij} denotes the synergy coefficient as $\Gamma_{ij} = \frac{\mathbf{S}_{ij}}{\mathbf{T}_{ij}}$, S_{ij} represents the total points scored when players

i and j share the court, T_{ij} indicates the total time both players are on the court together, $pos(i)$ signifies their total cooperative playing time during the game, $pos(i)$ denotes player positions, U_p is the maximum number of players at each position set by team strategy, and κ_p refers to the penalty intensity, set at 0.5 in this paper.

3.2.5. *Lineup Asset Value*

$$Asset(\mathbf{y}_t) = \sum_i V_i^{total} + Syn(\mathbf{y}_t) - Red(\mathbf{y}_t)$$

This article calculates the linear sum of individual values across each position. By evaluating the aggregated results for different positions, it identifies a team's strengths and weaknesses, providing a basis for determining the priority of player acquisitions.

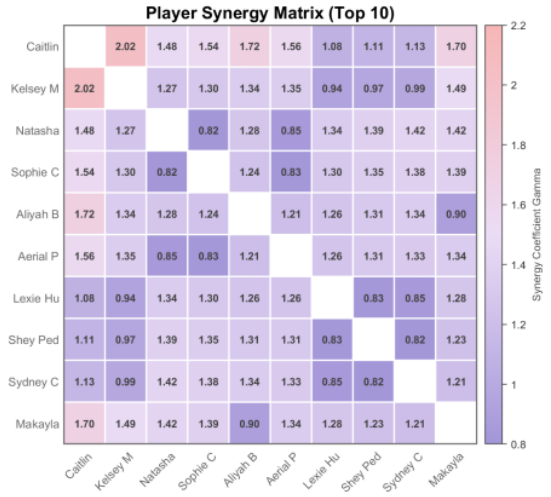


Figure 4: Synergy matrix

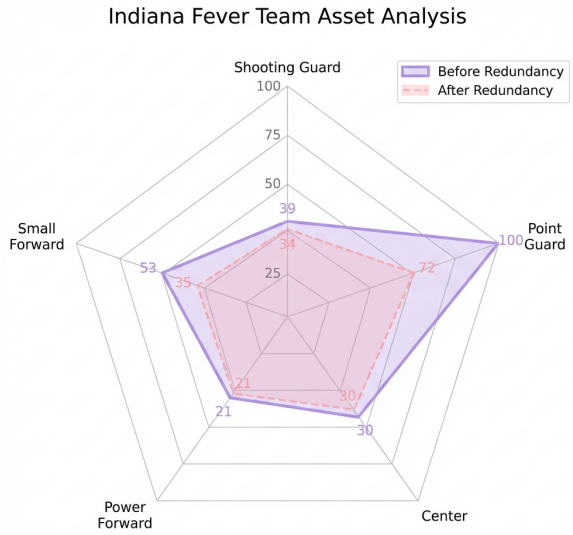


Figure 5: Radar chart of positional values

The left synergy matrix reveals a team synergy structure dominated by a few key combinations, where a small number of partnerships generate the primary amplification effect. The radar chart indicates that individual value is most prominent in the backcourt (especially point guards), but this value drops significantly when considering synergy redundancy, suggesting homogenous stacking and diminishing marginal returns. The frontcourt/interior dimension remains generally weak with minimal change before and after redundancy removal, suggesting the primary weakness stems from "absolute supply shortages" rather than redundant supply. Therefore, the team should stabilize high-synergy core combinations while prioritizing reinforcements for wing and interior roles to enhance structural balance and lower-bound stability.

3.2.6. *Player Acquisition*

This paper examines three distinct player acquisition scenarios in the WNBA, implementing differentiated personal value settings based on real-world data. Analysis reveals that draft picks typically possess high

potential but lower commercial value, while free agents command higher market value. Traded players generally demonstrate strong commercial and competitive performance within their teams.

Table 3: Player acquisition value formulas

Formula	Mode
$\mathrm{V}_{\text{total,draft}} = \alpha(1.5 \cdot V_{\text{perf}}) + (1-\alpha)(0.2 \cdot V_{\text{comm}})$	Draft
$\mathrm{V}_{\text{total,FA}} = \alpha(0.8 \cdot V_{\text{perf}}) + (1-\alpha)(1.2 \cdot V_{\text{comm}})$	**Free Agent Signings**
$\mathrm{V}_{\text{total,trade}} = \alpha V_{\text{perf}} + (1-\alpha)V_{\text{comm}}$	**Trades**

Based on the previously identified strengths and weaknesses, preliminary targets for new players are selected. Assuming the acquisition of this new player, they are incorporated into the overall model. By comparing the combined target $J(\cdot)$ from both inner and outer layers against the original lineup without this player, the resulting improvement in benefits is evaluated. These are then ranked, and the required number of players is selected.

$$\Delta J(k) = J(\mathbf{y} \cup \{k\}) - J(\mathbf{y})$$

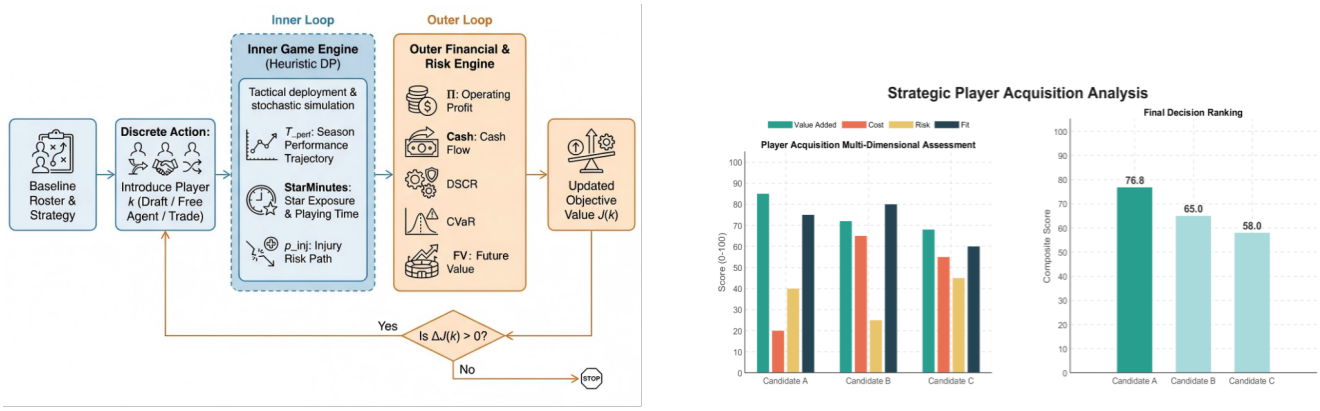


Figure 6: Player acquisition analysis

We simulated real-world data, and the output results shown in the figure on the right indicate that value and health are the primary criteria for recruiting new members.

3.3. Market Model

The model treats military expansion as an exogenous shock to the **oligopolistic market equilibrium**. Based on the impact of different expansion policies on the market, it introduces variations of the gravity model from spatial economics for quantification. Utilizing the **revenue dilution factor** δ_R and **market erosion factor** δ_M to influence the outermost targets w_1', w_2' *and* w_3', it generates new strategies according to market changes.

3.3.1. *Revenue Dilution Factor* δ_R

To reflect the decline in the proportion of broadcasting and sponsorship revenue allocated to each team due to league expansion, this paper introduces the revenue dilution factor:

$$\delta_R = \frac{1}{1 + \alpha_{dilute} \cdot \frac{\Delta N}{N}}$$

N represents the total number of teams before expansion, ΔN denotes the number of newly added teams, α_{dilute} is the dilution intensity coefficient, and indicates whether expansion "enlarges the pie." If $\alpha < 1$, it signifies incremental growth from new markets; if $\alpha \approx 1$, it represents a pure zero-sum game.

$$\ln(Rev_{avg}) = \beta_0 + \beta_1 \ln(N) + \beta_2 \text{TimeTrend} + \epsilon$$

$\alpha_{dilute} \approx 1 - \beta_1$, this paper takes into account short-term market stability by fixing **TimeTrend** to a constant value of 1, indicating no significant market trend changes during this period.

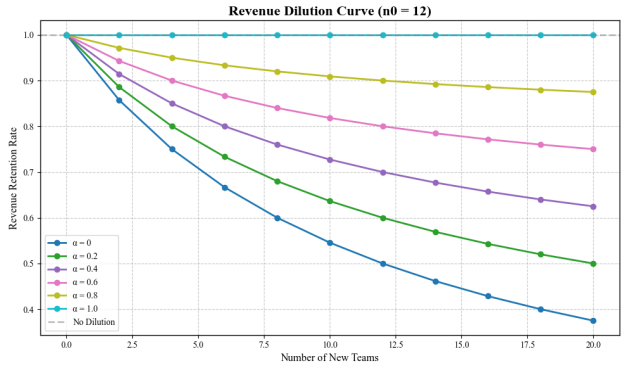


Figure 7: Average revenue retention rate

The figure illustrates the trajectory of the **average revenue retention rate** resulting from expansion across different dilution intensity coefficients as the number of new teams increases. When α_{dilute} is smaller, the curve declines more gradually, indicating that the new market generates stronger incremental revenue offsets and weaker dilution effects. When $\alpha_{dilute}\approx 1$, the decline approaches linearity with greater magnitude, corresponding to a near-**zero-sum game** where expansion primarily causes market share dilution; $\alpha_{dilute}=0$ equates to "zero dilution," serving as a benchmark.

3.3.2. *Market Erosion Factor* δ_M

This paper introduces a variant of the "gravity model" from spatial economics, using a dimensionless system to quantify how much potential market share new teams "steal" from existing teams by leveraging geographical advantages and market gravity[5].

$$\delta_M = 1 - \beta_{comp} \cdot \left(\frac{GDP_{new}}{GDP_{total}}\right)^{\theta} \cdot e^{-\lambda \cdot dist(i,new)}$$

$\frac{GDP_{new}}{GDP_{total}}$ is the gravitational parameter, controlling the impact of the relationship between the GDP of the new team and the total market GDP . In this paper, θ is set to 1, indicating a certain influence. $e^{-\lambda \cdot dist}$ is the distance parameter, showing that erosion increases exponentially as distance decreases. λ is the

spatial decay coefficient, typically set to $\lambda \approx 0.005 \sim 0.01(km^{-1})$ in sports economics. here set to 0.1 to indicate that market erosion gradually diminishes with increasing distance, though the rate of decay is not particularly rapid. $dist(i, new)$ is derived through latitude/longitude (ϕ, λ) and the **Haversine spherical distance formula**. β represents the conversion efficiency during market overlap, conservatively estimated at 0.5 in this study, indicating that two teams in the same city split the market evenly.

The figure above illustrates the potential erosion risk to the Indiana Fever market from major cities in the Midwest. This study finds Chicago to be the only expansion candidate city posing a significant threat, owing to its substantial economic scale and relatively close geographic proximity. In contrast, distant cities like Memphis exhibit erosion factors approaching zero, validating the effectiveness of the spatial decay effect.

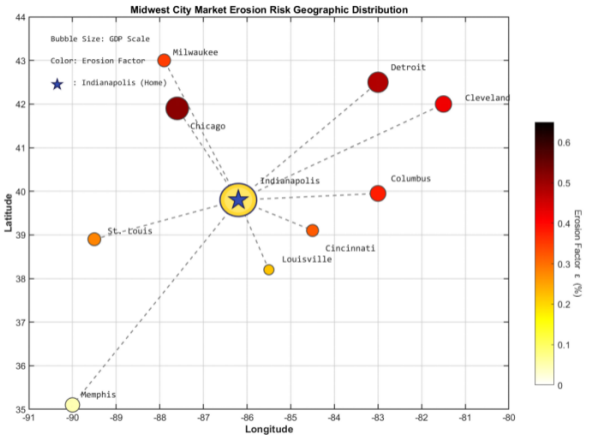


Figure 8: Market erosion risk for Indiana Fever

3.3.3. *Endogenous Drift in Risk Aversion*

When income is diluted$(\delta_R < 1)$, the cash flow safety cushion thins, and the threat of bankruptcy risk$(CVaR)$ rises nonlinearly. Management instinctively increases risk control weighting.

$$w_3' = w_3^{base} \cdot (1 + \eta_{risk} \cdot (1 - \delta_R)^{\gamma})$$

w_3^{base} represents the initial risk weight, γ is the nonlinear coefficient set to 2 in this paper, indicating that panic intensifies as losses approach the threshold, and η_{risk} denotes the **panic sensitivity**, which measures the aggressiveness with which management cuts expenses during revenue declines.

$$\eta_{risk} \approx \left| \frac{\%\Delta\text{Salary Cap Utilization}}{\%\Delta\text{League Revenue}} \right|_{\text{Crisis Year}}$$

Through historical data calibration against the impact of 2020, this paper sets $\eta_{risk} = 1.5$.

3.3.4. *Valuation Defense Drift*

When a new team enters a neighboring region$(\delta_M < 1)$, the existing market share faces threat. To preserve brand equity $(Brand_t)$, management places greater emphasis on long-term valuation (FV), meaning they are willing to spend heavily on advertising or acquire star players.

$$w_2' = w_2^{base} \cdot (1 + \eta_{defend} \cdot (1 - \delta_M))$$

η_{defend} represents **counter-sensitivity**.It estimates $\eta_{defend} \approx 0.3$ based on market calculations.

3.3.5. *Normalization*

$$w_1' = \max(\epsilon, 1 - w_2' - w_3')$$

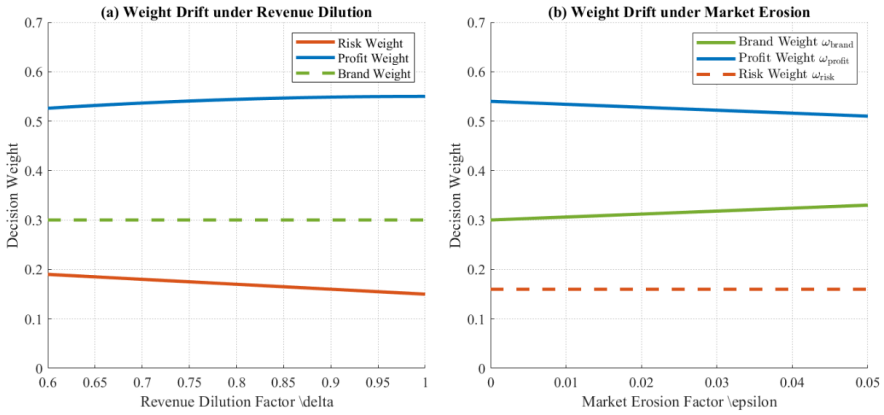


Figure 9: Weight adjustments under different market conditions

The left chart demonstrates that as income dilution intensifies ,when δ_R decreases,the cash flow safety cushion is compressed, and the tail bankruptcy risk captured by CVaR exhibits non-linear amplification. Consequently, the model reflects management's heightened risk control preference by increasing w_3, thereby correspondingly reducing the profit weighting under normalization constraints. The right chart illustrates that as market erosion intensifies—specifically when ϵ increases and δ_M decreases—the model elevates the valuation defense weight w_2 to sustain long-term valuation and brand equity. This reduces the short-term profit weight within the total weight constraint, while maintaining overall risk weights at relatively stable levels.

3.3.6. *Strategy Reconstruction*

Substitute the adjusted weights into the main decision model and solve for the optimal solution under the new environment.

$$y_{new}^*, \Delta D_{new}^* = \arg\max_{y,\Delta D} \mathbb{E}\left[\sum_{t=0}^{T} \frac{1}{(1+r)^t}(w_1'\Pi_t + w_2'\Delta FV_t - w_3' CVaR_\alpha)\right]$$

Results indicate that when δ_R is extremely low and η_{risk} is high, the model's output for $\sum s_{i,t} y_{i,t}$ (total payroll) drops significantly while ΔD_t (new debt) approaches zero. This suggests adopting a **consolidation strategy**: facing league-wide revenue dilution, the optimal solution is to cut premium players, hoard cash, and prevent $CVaR$ from skyrocketing. Conversely, when δ_M is extremely low (a new team is right next door), the model output $y_{i,t}$ favors high commercial value stars. Adopt a **moat strategy** sacrificing short-term profits by lowering prices and introducing superstars to defend market share and build a brand moat.

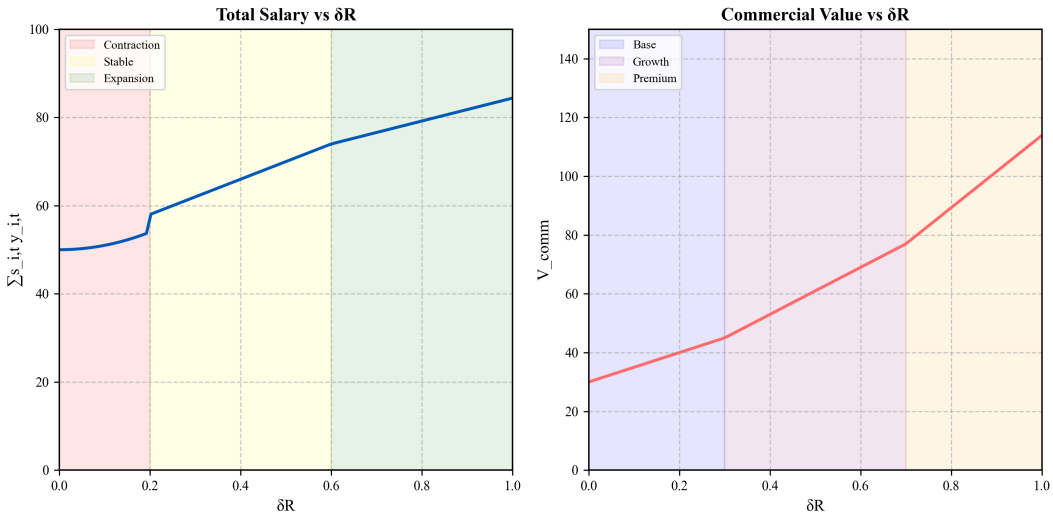


Figure 10: Strategy reconstruction results

3.4. Inner-Layer Game Model

To address the **"dimension catastrophe"** and **"nonlinear risks"** inherent in the WNBA's extended season (40 games), this paper constructs a "heuristic dynamic programming and Monte Carlo simulation system based on multidimensional constraint mapping."

3.4.1. *Player Attribute Parameters*

The results V^{total}, V_i^{perf} *and* V_i^{comm} from the value model in Section 3.2 are directly mapped.

- **Role Status** ($Role_i$), determines a player's tactical privileges, injury tolerance, and playing priority within the team, as ranked by their overall value V_i^{total} ranking.

$$Role_i = \begin{cases} 1.5 & \text{if } V_i^{total} \in \text{Top 10\% (Superstar)} \\ 1.2 & \text{if } V_i^{total} \in \text{Top 30\% (Starter)} \\ 1.0 & \text{if } V_i^{total} \in \text{Rotation (Rotation)} \\ 0.8 & \text{Otherwise (Bench)} \end{cases}$$

- **Effective Competitive Strength** (S_i^{eff}) indicates a player's marginal contribution to the team's win rate per unit of time.

$$S_i^{eff} = \gamma_1 \cdot V_i^{perf}$$

γ_1 is the dimensional adjustment coefficient. To ensure that the total S^{eff} of the entire team is on the same order of magnitude as the opponent's strength $OppStrength$, this paper sets γ_1 to 1.0.

- **Commercial Traffic Multiplier**(ϕ_i) indicates the multiplier effect of players on ticket prices and excitement levels, with the squared term reflecting the **"superstar effect"**.

$$\phi_i = \left(\frac{V_i^{comm}}{\text{Median}(V_{team}^{comm})} \right)^2$$

- **Position and Possession Characteristics**

Pos_i is derived from raw data $\{G, F, C\}$ and used for positional rigidity constraints. Actual usage rate (USG_i) is obtained from official data and used to calculate possession compatibility penalties.

3.4.2. *Decision Variables: Tactical Mapping and Dynamic Filling Algorithm*

- **Discrete Tactical Selection (a_g)** In the g match, the AI coach selects a tactical pattern:

$$a_g \in \{a_{\text{All-in}}, a_{\text{Standard}}, a_{\text{Rest}}\}$$

Tactical choices determine the **total time budget** for each position. Option $a_{\text{All-in}}$ reduces the rotation to 8 players, deploying core starters to their physical limits to maximize immediate win probability. Option a_{Standard} employs a standard 10-player rotation, balancing offense, defense, and stamina management. Option a_{Rest} utilizes deep bench players.

- **Tactics-Minute Mapping Matrix (T)**: Construct matrix $\mathbf{T} \in \mathbb{R}^{N\times 3}$ to pre-store minute templates for three modes. Upon selecting a_g, the **base minute vector** $\mathbf{m}_g^{base}$ is immediately determined:

$$\mathbf{m}_g^{base} = \mathbf{T} \cdot \mathbf{1}_{a_g}$$

Matrix $\mathbf{T} \in \mathbb{R}^{N\times 3}$ serves as the model's static database, generated entirely automatically based on the role status ($Role_i$) defined by nodes.

Table 5: Minute allocation by role and tactical mode

$Role_i$	$\mathbf{v}_{All-in}$	$\mathbf{v}_{Standard}$	$\mathbf{v}_{Rest}$
Superstar ($Role = 1.5$**)**	40mins	32mins	0or15mins
Starter ($Role = 1.2$**)**	35mins	28mins	20mins
Rotation ($Role = 1.0$**)**	15mins	18mins	25mins
Bench ($Role = 0.8$**)**	0mins	5mins	20mins

- **Depth Chart Dynamic Sorting:** Before each game, sort players by position based on their current **health status** $h_{i,g}$ and V^{total} **rank**: **Guard List**: $\{G_1, G_2, \dots\}$ (sorted in descending order by $V^{total} \cdot h$), **Forward List**: $\{F_1, F_2, \dots\}$, **Center List**: $\{C_1, C_2, \dots\}$. The model fills the time budget prioritizing by rank until 240 minutes are allocated. If top-ranked players exhaust the time, lower-ranked players automatically receive **0** minutes.

$$m_{i,g} = \min(\text{Remaining Position Budget},\ m_{i,g}^{\max})$$

3.4.3. *Constraints*

- **Based on a health-based dynamic upper limit, a time limit has been established.**

$$m_{i,g} = \min(\text{Remaining Position Budget},\ \text{Personal Stamina Limit}\ m_{i,g}^{\max})$$

$$\rho_{i,g} = \frac{h_{i,g} \cdot Role_i}{\sum_j (h_{j,g} \cdot Role_j)}$$

$Role_i$ is determined by V^{total}, meaning that for a superstar of V^{total} stature, even with a minor injury, the denominator weight remains significant—reflecting the privilege of playing through injury.

- **Positional rigidity constraints ensure** the structural balance of basketball games:

$$\sum_G m_{i,g} \approx 96 \quad \sum_F m_{i,g} \approx 96 \quad \sum_C m_{i,g} \approx 48$$

- **Ball-handling compatibility penalty: In practice,** when the combined true usage rate (USG) of all five players on the court is excessively high, efficiency suffers. This paper introduces a penalty coefficient η_{usage} when calculating win probability.

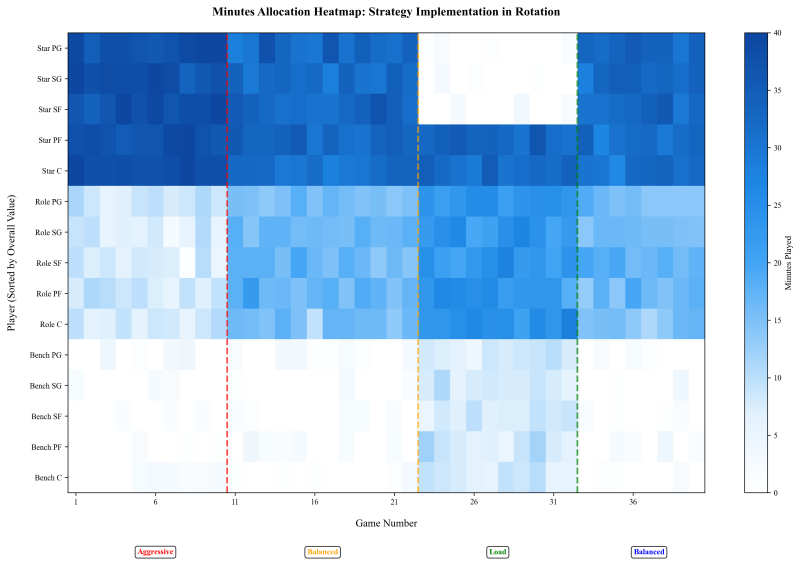


Figure 11: Minute allocation across tactical phases

This article directly maps the strategy sequence to rotations. The diagram above shows that during the Balanced phase, the main players maintain stable minutes with only minor fluctuations. Upon entering

the Aggressive phase, the overall color of the main players deepens, reflecting the intensified core lineup. When switching to the Load phase, the core lineup cools significantly, even experiencing low minutes or blank slots, while role players and the bench area heat up and take over. Finally, the rotation returns to Balanced, resuming the normal configuration dominated by stable starters.

3.4.4. *Risk Engine*

This paper utilizes the Instreet injury database and employs a Logit survival analysis model to calculate the **injury risk coefficient vector ($\mathbf{\Lambda}$)**.

$$P(\text{Injury}) = \text{Sigmoid}(\lambda_0 + \lambda_1 \cdot \text{Minutes} + \lambda_2 \cdot \text{Age} + \lambda_3 \cdot \text{RestDays})$$

Smooth randomness through 200 Monte Carlo simulations. **Random injury generation** calculates risk probability $p_{i,g}^{inj}$ using coefficient $\mathbf{\Lambda}$, then performs a Bernoulli trial:

$$I_{i,g}^{(k)} \sim \text{Bernoulli}(p_{i,g}^{inj})$$

Based on simulations, evolve health status and statistically output: Take the average value $\bar{h}_{i,g}$ from K number of simulations.

$$h_{i,g+1}^{(k)} = \text{Clip}\left(h_{i,g}^{(k)} + \rho_{rec}(1 - \mathbb{I}_{Match}) - \xi m_{i,g} - \chi I_{i,g}^{(k)},\ 0,\ 1\right)$$

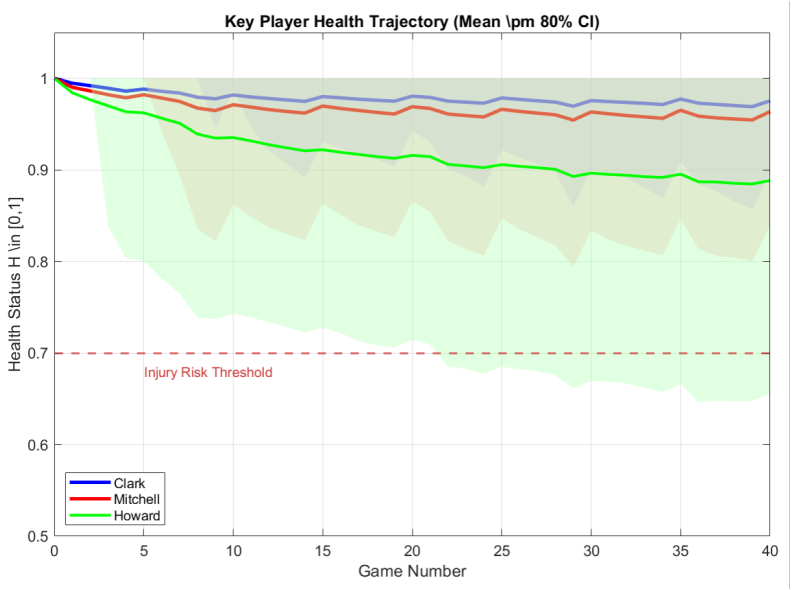


Figure 12: Health simulation results

The results indicate that, based on our simulation findings, the core player curve exhibits overall greater stability with a narrower range and remains farther from the threshold. This demonstrates that the strategy mitigates potential nonlinear health declines associated with Aggressive through load reduction arrangements such as Load, enabling the model to achieve simultaneous management of competitiveness.

3.4.5. *Quantifying Output*

Introducing ball possession penalties to adjust lineup strength ($Perf_g$)

$$Perf_g = \eta_{usage} \cdot \sum_{i=1}^{N} (S_i^{eff} \cdot m_{i,g}^{final})$$

Among them, the ball possession penalty coefficient:

$$\eta_{usage} = \exp(-\lambda_{usg} \cdot \max(0, \bar{USG}_{oncourt} - 100\%))$$

If the court is filled with ball-dominant players ($\sum USG > 100\%$), η will drop below 1, preventing the team from reaching its full potential.

- **Single-Game Win Rate Stream($WinProb_g$)**

$$WinProb_g = \text{Sigmoid}(\kappa \cdot (Perf_g - \text{OppStrength}_g))$$

κ represents win probability sensitivity, calculated in this paper via logistic regression: Win $\sim$ (TeamRating - OppRating).Specifically, for every 1-point increase in the team rating difference, the log-odds of winning increases by 0.15.

- **Single-Game Excitement Flow($Experience_g$)**

$$Experience_g = \sum_{i=1}^{N} \phi_i \cdot \ln(m_{i,g}^{final} + 1)$$

- **LTV** dynamically contributes points across 40 matches to support subsequent season tickets.

$$LTV_i = \underbrace{w_1 \sum_{g=1}^{40} \left(\bar{WinProb}_g \cdot \frac{\bar{m}_{i,g}}{240} \right)}_{\text{Wins}} + \underbrace{w_2 \sum_{g=1}^{40} \left(\bar{Experience}_g \cdot \frac{\bar{m}_{i,g}}{240} \right)}_{\text{Stars}} + \underbrace{w_3 \cdot \frac{1}{40} \sum_{g=1}^{40} P(h_{i,g} > 0.6)}_{\text{Reliability}}$$

3.4.6. *Solution Algorithm: Heuristic ADP[3]*

To find the optimal tactical sequence $\{a_1^*, ..., a_{82}^*\}$, we employ **Heuristic ADP to construct the Bellman equation:**

$$J(S_g) = \max_{a \in \mathcal{A}} [\mathcal{R}(WinProb_g, Experience_g) + \gamma \cdot \mathbb{E}[J(S_{g+1}) \mid S_g, a]]$$

$\mathcal{R}$ represents immediate financial returns (Ticket Revenue), $\mathbb{E}[J]$ denotes future expected value (including LTV retention), then apply a **value function approximation** ,using linear basis functions to approximate future value, thereby resolving the state explosion problem:

$$\widetilde{J}(S_{g+1}) \approx \omega_1 \cdot \sum h_{i,g+1} - \omega_2 \cdot \sum_{Stars} (1-h_{i,g+1})^2$$

The model learned **"fear"** through component ω_2—if overusing star players causes their health to decline, their future value $\widetilde{J}$ will plummet. This forces the AI to proactively choose a_{Rest} for rest rotations. In Monte Carlo simulations, **TD-Learning** updates weights ω using the temporal difference error (TD-Error) until the policy converges.

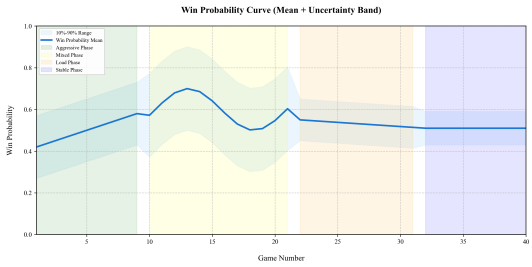


Figure 13: Win probability with uncertainty bands

The figure above shows the mean curve of win rate per game alongside the 10%–90% percentile uncertainty band. During periods of strong opponents or intense schedules, the mean fluctuates or declines while the band width increases. Comparing different strategy phases reveals that the Aggressive phase exhibits a higher mean but broader uncertainty, while the Load phase shows a slightly lower mean yet greater stability with a narrower band.

3.5. Financial Module

This model converts $WinProb$, $Experience_g$, and LTV into revenue and profit, directly forming DSCR constraints and risk inputs,thereby establishing connectivity between the inner and outer models.

3.5.1. *Season Ticket Revenue*

To enable the model to "adapt to market conditions," we artificially constructed a virtual team of 15 "standard players", $Role = 1.0$, with all ability values set to the median of the sample—and substituted them into the previous model. The total value points earned after 40 games serve as the benchmark value points (LTV_{bench}), while introducing the market equilibrium pricing coefficient (η_{mkt}) represents the premium or discount multiplier.

$$\eta_{mkt} = 0.85 \cdot \left(\frac{\mathbf{LTV}_{team}}{LTV_{bench}} \right)^{\gamma}$$

The exponent term γ=0.6 represents the sensitivity of prices to fluctuations in team value. Based on the average consumption threshold base ticket price P_{base}=100$ derived from the WNBA's median

market, the total unit price P_{season} for season tickets is no longer static. Instead, it is dynamically generated according to the "value" demonstrated by the team in the simulation:

$$P_{season} = \eta_{mkt} \cdot \left(\sum_{g=1}^{40} P_{base,g} \right)$$

Season ticket sales N_{ST} depend on fans' overall assessment of "value for money"—the interplay between the appeal of the team's star power and the deterrent of ticket prices.

$$N_{ST} = \text{MarketSize} \cdot \left(k_1 \cdot \ln \left(\frac{\mathbf{LTV}_{team}}{LTV_{bench}} \right) - k_2 \cdot \eta_{mkt} + C \right)$$

This study employs the boundary calibration method to determine k_1= 0.52, indicating that a 1% increase in brand equity score positively drives sales volume. The logarithmic representation reflects diminishing marginal utility.k_2=0.15 signifies that a higher pricing coefficient η_{mkt} correlates with lower purchase intent, representing the team's core fan base.

3.5.2. *Single-Game Ticket Revenue*

Based on excitement for each match, ticket prices for each game will fluctuate above the base price P_{base}.

$$P_g = P_{base} \cdot \bar{Experience}_g$$

$$N_{single,g} = \min \left(C_{remain}, D_{base} \cdot \left(\frac{P_g}{P_{base}} \right)^{-\eta} \cdot (\bar{Experience}_g)^{\phi} \right)$$

$N_{single,g}$ represents individual ticket sales volume,C_{remain} denotes the stadium's remaining capacity ($Capacity - N_{ST}$), η is price elasticity of demand (typically $\eta > 1$), reflecting fans' sensitivity to price increases—the model assumes a value of 1.5, ϕ is quality sensitivity, measuring the impact of match quality on purchasing intent—assumed to be 1.0.

The results reveal that ticket prices exhibit the highest upward deviation under the Aggressive strategy, moderate deviation under Balanced, and overall lower, smoother deviation under Load. These three curves reflect the trade-off paths between revenue growth through price increases and demand decline under different pricing strategies.

3.5.3. *Total Revenue*

$$R_t = R^{ticket} + R^{broadcast} + R^{endorsement}$$

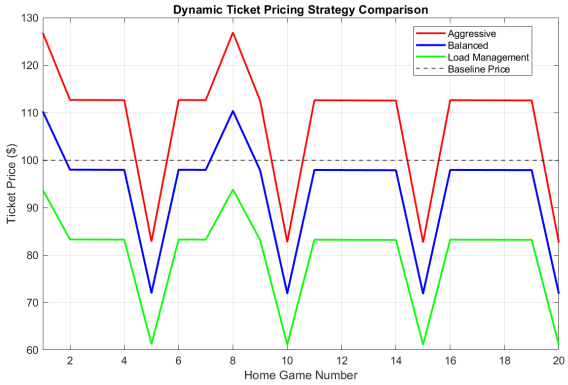


Figure 14: Ticket price deviations under different strategies

$$R_{ticket} = N_{ST} \cdot P_{season} + \sum_{g=1}^{40} (N_{single,g} \cdot P_g) \qquad R^{broadcast} = \phi_{broadcast} \times \text{Market}_t$$

Broadcast revenue $R^{broadcast}$ is related to broadcasting rights and broadcast fees for events. This paper represents it using market size and brand influence. $\phi_{broadcast}$ is the coefficient for broadcast revenue, obtained through regression analysis of historical data, broadcasting agreements, and broadcast rights. Market_t indicates the degree of influence market size has on broadcast revenue. Market size is calculated using the following formula:

$$\text{Market}_t = w_1 \times GDP_t + w_2 \times Population_t$$

GDP_t represents the economic scale of the region, $Population_t$ reflects market potential and purchasing power, while weights w_1 and w_2 are derived through regression analysis to indicate the relative importance of GDP and population on market size.

$$R^{endorsement} = \phi_{endorsement} \times Brand_t \times Market_t$$

Endorsement income is jointly determined by brand influence and market size. $\phi_{endorsement}$ represents the degree of influence of brand influence and market size on endorsement income. This paper obtained through regression analysis that $Brand_t$ represents brand influence, quantified by the indicators of **brand loyalty** and **brand awareness**:

$$Brand_t = \alpha_1 \cdot \text{Brand Awareness}_t + \alpha_2 \cdot \text{Brand Loyalty}_t$$

Quantify $BrandLoyalty_t$ by the ratio of season ticket holders to venue capacity.Reflected $BrandAwareness_t$ through **social media follower counts**:

$$\text{Brand Loyalty}_t = \frac{\text{Season Ticket Holders}_t}{\text{Average Venue Capacity}} \qquad \text{Brand Awareness}_t = \frac{\text{Social Media Followers}_t}{\text{Max Social Media Followers}}$$

Max Social Media Followers represents the highest number of followers within a specific timeframe or target market. Based on industry experience, brand awareness contributes significantly to brand influence, hence setting α_1 to 0.6 and α_2 to 0.4.

3.5.4. *Operations and Financing*

$$C_t^{salary} = \sum_i s_{i,t} y_{i,t} \qquad Tax_t = \tau \cdot \max(0, C_t^{salary} - C_{cap,t})$$

$$C_t^{op} = C_t^{coach} + C_t^{train} + C_t^{medical} + C_t^{venue} \qquad \Pi_t = R_t - C_t^{salary} - C_t^{op} - Tax_t - i_t D_t$$

$s_{i,t}$ represents Spotrac salaries, constrained by the WNBA salary cap; $C_{cap,t}$ denotes the luxury tax rate;i_t indicates the loan interest rate; C_t^{op} is the cost of capital derived from historical averages. Profit consolidates roster salaries, financing interest, and taxes into a cash flow metric, serving as input for subsequent DSCR/CVaR calculations.

$$D_{t+1} = (1 + i_t)D_t + \Delta D_t - Rep_t \qquad Cash_{t+1} = Cash_t + \Pi_t - \text{CapEx}_t + \Delta D_t + \Delta E_t$$

$\Delta D_t, Rep_t, \Delta E_t$ represents the outer-layer decision variable, CapEx_t denotes the fixed value of capital expenditure, and c constitutes the core of the closed-loop feasibility domain from financing to lineup.

3.5.5. *Implementation of Financial Risk Control*

To better serve the team, the model incorporates the debt coverage ratio $DSCR_t$ from economics and finance, along with the conditional value at risk (CVaR) for measuring tail risk in investment portfolios.

$$EBITDA_t \approx R_t - C_t^{salary} - C_t^{op} - Tax_t \qquad DebtService_t = i_t D_t + Rep_t$$

$$DSCR_t = \frac{EBITDA_t}{DebtService_t} \geq DSCR^*$$

$DSCR^*$ [5] represents the safety threshold, a **contractually hard constraint** set at 1.3 in this paper according to industry regulations. To incorporate financial dynamic risk monitoring into portfolio

optimization, CVaR is introduced—the average expected loss when losses exceed VaR (Value at Risk) at a given confidence level.

$$L_{t,s} = \max(0, Cash^{\min} - Cash_{t,s}) \qquad z_{t,s} \geq L_{t,s} - \eta_t,\ z_{t,s} \geq 0$$

$$CVaR_\alpha(L_t) = \eta_t + \frac{1}{(1-\alpha)S} \sum_{s=1}^{S} z_{t,s}$$

$\mathbf{L_t}$ represents **cash flow loss**, while $\text{Cash}^{\min}$ denotes the minimum cash flow a company must maintain to avoid bankruptcy or liquidity risk. This paper sets b as the sum of the company's three-month costs. Using the value calculated by our model as the reference standard, we then employ Monte Carlo simulation to generate multiple potential **scenarios** representing varying economic conditions that evolve throughout the simulation. **VaR**(η_t) represents the **maximum loss** at a given confidence level, calculated from all simulation results. Finally, **CVaR** is derived by taking the weighted average of all losses exceeding the **VaR** threshold.

This chart conducts a sensitivity analysis across different confidence levels. As the confidence level increases, the risk metric rises significantly. Moreover, the average loss in the worst-case scenario consistently exceeds the quantile loss, with the gap between the two gradually widening. This indicates that cash flow gaps in extreme scenarios exhibit a stronger amplification effect. When management opts for a more conservative confidence level, the model places greater emphasis on monitoring and constraining risks associated with a small number of worst-case economic scenarios, rather than focusing solely on losses within the typical fluctuation range.

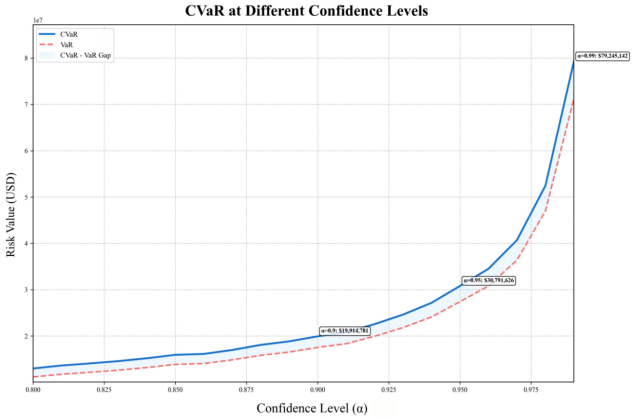


Figure 15: CVaR sensitivity analysis

3.6. Potential for Future Development ΔFV_t

$$FV_t = \psi \cdot R_t + \phi_{endorsement} \times Brand_t \times Market_t + \phi_{broadcast} \times \text{Market}_t$$

In this paper, ψ is set to 1, while the remaining quantities are derived from prior calculations. The variable FV_t measures the company's overall value at a specific point in time t, indicating its potential for future development. ΔFV_t represents the temporal variation of the outer-layer overall objective FV_t.

$$\Delta FV_t = FV_t - FV_{t-1}$$

Compared to short-term profits,ΔFV_t focuses more on long-term value changes and helps decision-makers identify growth opportunities or potential risks.

3.7. Outer Layer Overall Objective

$$\max \mathbb{E}\left[\sum_{t=0}^{T} \frac{1}{(1+r)^t}(w_1\Pi_t + w_2\Delta FV_t - w_3 CVaR_\alpha(L_t))\right]$$

All models are interconnected and integrated, forming a dynamic decision-making framework that spans the entire season. This enables management to simultaneously adjust roster structure and on-field performance while comprehensively evaluating risk and cash flow, thereby maximizing team profitability and long-term value. It empowers team owners and general managers to dynamically adjust leverage based on shifts in team performance and financial conditions.[5]

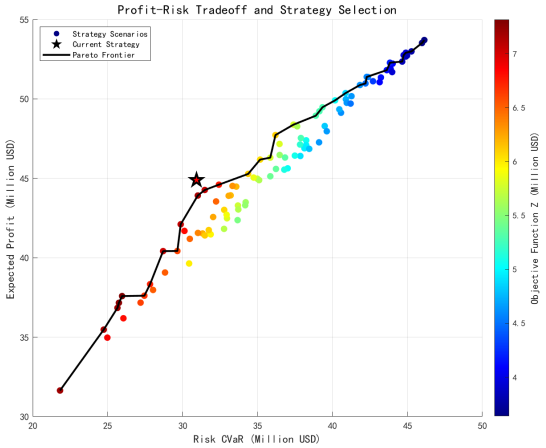


Figure 16: Risk-return analysis with Pareto frontier

This chart maps different strategy scenarios onto a risk-return plane, with the overall scatter plot trending toward the upper right. This indicates that under this model, higher returns typically come with greater tail risk. The black dashed line represents the "Pareto frontier,"[4] which denotes a set of optimal trade-offs offering the highest returns for a given level of risk, or the lowest risk for a given level of return. The asterisk marks the current strategy position. Management's decision involves selecting different points along the frontier based on risk preferences.

4. Sensitivity Analysis and Robustness Testing

Core players represent crucial parameters within the model. Therefore, we conduct sensitivity analysis on their playing status. When a core player sustains an injury in the gth game, the model treats this as a **state update**: the player's health decreases, and the optimal strategy is resolved anew.

Core player injuries are modeled as health status impacts. The model recalculates tactical patterns and minute allocation for the remaining schedule based on updated health statuses, illustrating the adjustment logic of "Load insertion + concentrated effort in key matches" through strategy comparison

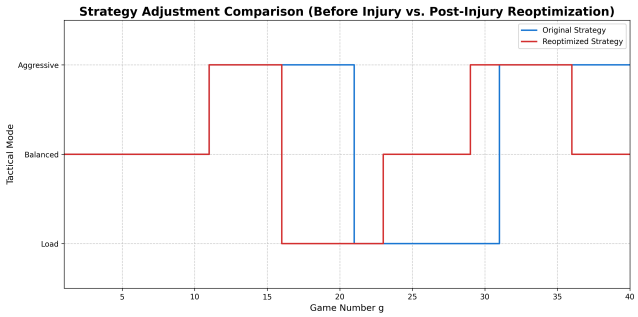


Figure 17: Strategy adjustment

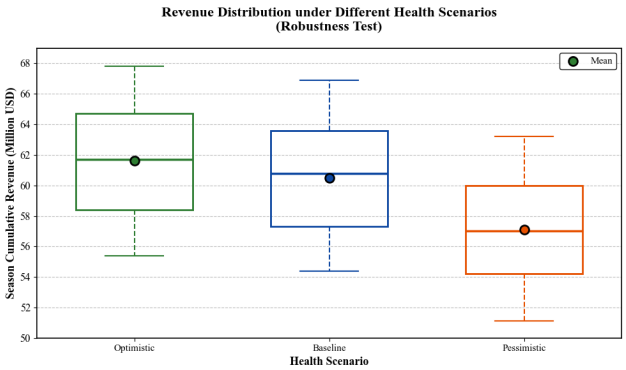


Figure 18: Robustness testing

charts.Furthermore, robust testing was conducted on the distribution of core metrics such as wins and revenue under varying injury severity scenarios (initial health/impact magnitude). Results confirm that the optimal contingency strategy maintains its advantage across multiple scenarios, validating the model's stability under injury uncertainty.

5. Model Evaluation

5.1. Advantages

An integrated inner–outer framework converts management levers into season strategies and uses Monte Carlo + CVaR to deliver robust, risk-aware recommendations.

5.2. Disadvantages:

Performance is highly sensitive to parameter/data quality and to exogenous inputs (e.g., schedule and opponent strength), so estimation errors can propagate into win-probability and strategy outputs.

References

[1] Chow Y, Ghavamzadeh M. Algorithms for CVaR optimization in MDPs[J]. Advances in neural information processing systems, 2014, 27.

[2] Zhu Y, Tian D, Yan F. Effectiveness of entropy weight method in decision-making[J]. Mathematical problems in Engineering, 2020, 2020(1): 3564835.

[3] Aghazadeh D, Ertogral K. An improved approximate dynamic programming method for the Integrated Fleet Sizing and Replenishment Planning Problem with predetermined delivery frequencies[J]. IFAC-PapersOnLine, 2022, 55(10): 3034–3039.

[4] Van Den Berg E, Friedlander M P. Probing the Pareto frontier for basis pursuit solutions[J]. Siam journal on scientific computing, 2009, 31(2): 890–912.

[5] Google Gemini (January 2026 version, Gemini3.0)



MEMO
TO: Indiana Fever Owner and General Manager
FROM: Team #2628552
DATE: February 2, 2026
SUBJECT: Strategic Briefing: Building a Long-Term Moat Where Wins Generate Profits

Dear Management:

To unify competitive pursuit with financial returns, we have developed an "integrated inner-outer layer" decision-making system for you. This system eliminates the disconnect between tactics and finances by mapping on-court performance in real-time to balance sheet figures. Under continuous monitoring of the financial safety cushion, it dynamically adjusts game strategies to ensure every minute of investment serves long-term valuation.



Regarding roster construction, data has issued a clear warning: the commercial and competitive value derived from backcourt anchors has reached saturation. Further stacking guards will face severe diminishing returns. We strongly recommend shifting the recruitment focus entirely toward forwards and centers. Introducing a potent frontcourt not only provides tactical structural complementarity but is also essential to prevent excessive wear and tear on backcourt anchors, safeguarding the team's most valuable assets.



Facing inevitable league expansion, particularly the potential threat from a new Chicago franchise, we must implement a proactive "moat strategy." This entails strategically accepting short-term profit compression during the expansion phase, locking in fan loyalty through high-caliber signings and flexible ticket pricing incentives. Today's strategic investments are essentially "insurance" purchased to defend against future market share erosion.



To ensure the security of this strategy's implementation, we have established two bottom lines: First, implementing "dynamic pricing" to adjust ticket prices in real-time based on game popularity, maximizing revenue. Second, establishing an "injury circuit breaker mechanism" that immediately triggers load management mode upon any health warning for core players. We would rather sacrifice short-term regular-season win rates to guarantee the integrity of our core roster during the playoffs.

This solution aims to harness rational quantitative models to navigate the emotional landscape of competitive sports, helping you remain undefeated on both the commercial and athletic fronts.

Yours sincerely,

Team #2628552

Report on Use of AI Tools

1 DeepL SE, DeepL Translator (January 2026 version)

Used for full-text translation; final translations undergo human review and adjustment to ensure accuracy.

2. Google *Gemini* (January 2026 version, Gemini3.0)

Query: Please explain the concept of the "inner-outer loop framework" in sports team management and how it can be applied to balance competitive performance with financial profitability.

Output: The inner-outer loop framework is a hierarchical decision-making architecture.The outer loop operates at the macro-strategic level, focusing on long-term valuation optimization, financial risk management (DSCR constraints and CVaR), resource allocation decisions, and market response modeling. The inner loop operates at the tactical level, encompassing playing time allocation, tactical pattern selection, health status tracking, and performance metric generation. Both layers interact via bidirectional data flow: the outer layer sets constraints and objectives, while the inner layer simulates outcomes of various tactical scenarios and feeds results back to update financial projections and risk assessments.

Adoption Status: We adopted the hierarchical framework describing outer and inner loops but independently developed the mathematical formalization of the objective function, constraint set, and Monte Carlo simulation coupling mechanism.

3. Google *Gemini* (January 2026 version, Gemini3.0)

Query: Compare and analyze different player valuation methods in professional sports, including the strengths and weaknesses of performance metrics and commercial value metrics.

Output: Player valuation methods include: (1) Traditional technical statistics—easily accessible but insufficiently capture defensive impact; (2) Advanced analytical metrics (PER, WS, BPM)—more comprehensive but computationally complex; (3) Tracking data and spatial analysis—capture off-ball contributions but limited historical data;(4) Commercial value assessments—directly measure revenue-generating potential but exhibit high volatility; (5) Hybrid dual-track valuation systems—comprehensive but subject to subjective weighting. We recommend using entropy weighting to determine objective weights, z-score standardizing metrics, and incorporating synergy and redundancy adjustments.

Adoption Status: We adopted the dual-track value stream framework and entropy weighting recommendations, but independently developed the specific metric set, synergy coefficient formula, and variance coefficient calibration method.

4. Google *Gemini* (January 2026 version, Gemini3.0)

Query: Explain the gravity model in spatial economics and how it analyzes market competition and revenue erosion between sports teams in different cities.

Output: The gravity model assumes interactions between two locations are proportional to their economic quality and inversely proportional to distance. Applied to sports markets, the market erosion factor formula is: $E = \beta \cdot (GDP_{new}/GDP_{market})^{\alpha} \cdot e^{-\lambda \cdot d}$. Key parameters include: distance decay coefficient λ (typically 0.05–0.2 in sports economics), gravity parameter α, and conversion efficiency β (approx. 0.4–0.6 for same-city, 0.1–0.3 for adjacent markets).

Adoption Status: We adopted the gravity model structure and spatial decay concept but independently specified all parameter values. We calculated actual distances using the Haversine formula and created original visualizations of erosion risk for Midwestern cities.

---

5. Google *Gemini* (January 2026 version, Gemini3.0)

Query: Explain heuristic approximate dynamic programming (ADP) and its application in allocating playing time for sports teams during a season.

Output: ADP solves large-scale sequential decision problems by approximating the value function. Core components include: the Bellman equation ($V(s_t) = \max_{a_t}\{r(s_t, a_t) + \gamma \mathbb{E}[V(s_{t+1})]\}$); value function approximation ($\tilde{V}(s;\theta) = \sum_k \theta_k \phi_k(s)$), with base functions encompassing team health, star player availability, remaining games, etc.; TD learning updates weights. Heuristic components include state aggregation, action space reduction (aggressive/balanced/load management modes), and policy expansion.

Adoption: We adopted the general ADP framework while independently specifying WNBA-specific base functions and tactical mode mapping matrices. We calibrated injury probability models using InStreet data.

---

6. Google *Gemini* (January 2026 version, Gemini3.0)

Query: Explain Conditional Value at Risk (CVaR) and its application advantages in sports team financial management.

Output: CVaR is defined as the expected loss when losses exceed VaR:$CVaR_{\alpha} = \mathbb{E}[L \mid L \geq VaR_{\alpha}]$. Calculation steps: Define loss variables, generate loss distributions via Monte Carlo simulation, compute VaR quantiles, and average losses exceeding VaR. Compared to standard deviation (symmetrically treating upside/downside) and VaR (ignoring tail severity), CVaR focuses on tail risk, satisfies coherence, is optimization-friendly, and is more conservative.

Adoption: We adopted the CVaR framework and computational procedures, but independently defined the loss variable in the WNBA context, designed the Monte Carlo simulation structure, specified the confidence level (α=0.95), and created sensitivity analyses.

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7. Google *Gemini* (January 2026 version, Gemini3.0)

Query: How can sports teams optimize season ticket pricing dynamically? What factors should be considered?

Output: Demand factors include: team performance, star player availability, opponent appeal, timing, and economic conditions. Modeling methods include: demand curve estimation ($Q = f(P, X)$), price elasticity (typically 0.5-2.0 for sports), and revenue optimization. Dynamic considerations include intertemporal optimization and state-dependent pricing. Multi-objective

frameworks balance revenue, attendance, and lifetime value.

Implementation Approach: We adopted the general dynamic pricing framework but independently developed a model linking match entertainment value to inner-layer simulation outputs, creating a price adjustment formula based on tactical patterns.

8. Google *Gemini* (January 2026 version, Gemini3.0)

Query: How to use the Entropy Weighting Method to determine objective weights for multi-criteria decisions in player evaluations?

Output: Entropy weighting steps: (1) Construct decision matrix; (2) Standardize to [0,1]; (3) Calculate probability matrix$p_{ij} = r_{ij}/\sum r_{ij}$; (4) Calculate entropy$e_j = -\frac{1}{\ln m}\sum p_{ij} \ln\, p_{ij}$; (5) Differentiation degree$d_j = 1 - e_j$; (6) Weight$w_j = d_j/\sum d_k$. Advantages include objectivity, adaptability, focus on discriminative capability, and transparency.

Adoption Status: We fully adopted the mathematical formula of the entropy weight method but added z-score preprocessing, supplemented α parameter calibration with the coefficient of variation method, and incorporated collaborative redundancy adjustment.

9. Google *Gemini* (January 2026 version, Gemini3.0)

Query: Explain Debt Service Coverage Ratio (DSCR) and its application as a financial covenant constraint for sports teams.

Output: DSCR = Net Operating Revenue / Total Debt Service. Threshold interpretation: <1.0 indicates insufficient coverage, 1.2–1.5 adequate, >2.0 strong coverage. Sports-specific considerations include revenue volatility (requiring higher buffers), seasonality, and salary cap dynamics. Constraint in optimization:$NOI(\vec{x}) \geq DSCR_{min} \times R$.

Adoption Status: We adopted the DSCR definition and 1.30x threshold, but independently specified revenue-cost components for the WNBA financial module and linked DSCR to the CVaR framework.

10. Google *Gemini* (January 2026 version, Gemini3.0)

Query: How can synergistic and redundancy effects within sports team rosters be modeled?

Output Content: Synergy Coefficient MatrixS_{ij} Measures the degree to which players amplify each other's performance (>0 indicates complementarity, <0 indicates redundancy). Position redundancy penalty uses a quadratic form: $\sum_p \lambda_p \cdot \max\,(0, n_p - n_p^*)^2$. Usage rate compatibility penalty accounts for conflicts among players with high ball-handling demands. Total Roster Value = Sum of Individual Values + Synergy Effects - Redundancy Penalty.

Adoption Status: We adopted the synergy matrix and redundancy penalty framework, but independently developed the synergy coefficient formula based on WNBA data, setting λ=0.5. We created heatmaps and radar charts for visualization.

11. Google *Gemini* (January 2026 version, Gemini3.0)

Query: How does league expansion economically impact existing sports teams?

Output: Impact mechanisms include: broadcast revenue dilution (dilution factor = N/(N+ΔN)), market overlap and fan erosion, competitive balance effects. Defensive strategies include strengthening local markets and securing long-term commitments. Favorable expansion locations are distant large markets; detrimental locations are adjacent overlapping markets. Historical data shows local revenue declines of 5-15% for teams within 100 miles.

Adoption Status: We adopted the expansion impact analysis framework but independently developed the revenue dilution formula with δ parameter, calibrated gravity model parameters, created visualizations, and designed an endogenous weight drift mechanism alongside "moat strategy" recommendations.

12. Google *Gemini* (January 2026 version, Gemini3.0)

Query: Optimize the sentence "When key players are injured, we need to change strategies and give other players more playing time."

Output: The professional version includes: "Key player injuries are modeled as exogenous shocks to the health state vector, prompting the model to re-solve the remaining schedule optimization problem with updated constraints." and "The decision framework initiates contingency protocols, adjusting playing time allocations to balance competitive performance with health management considerations."

Adoption Status: We adopted the concepts of "exogenous shocks" and "resolving optimization problems" and integrated them with our model terminology.

13. Google *Gemini* (January 2026 version, Gemini3.0)

Query: What is the Pareto frontier, and how can it be used to analyze risk-reward tradeoffs for sports teams?

Output: The Pareto frontier represents a set of solutions where no single objective can be improved without worsening another. Generation methods include weighted sum and ε-constrained approaches. Conservative strategies (lower left) on the frontier correspond to low-risk, low-reward outcomes, while aggressive strategies (upper right) indicate high-risk, high-reward scenarios. Slope values denote marginal substitution rates, guiding management to select operating points based on risk preferences.

Adoption Status: We adopted the Pareto frontier concept but independently generated the frontier via Monte Carlo simulation. We created a scatter plot specific to the Indiana Fever and developed a framework for management recommendations.

14 Google *Gemini* (January 2026 version, Gemini3.0)

Query: How to interpret and present sensitivity analysis results for key player injuries?

Output: Analysis types include single-parameter sensitivity, scenario analysis, and Monte Carlo sensitivity. Health impact formula: $h^{后} = h^{前} - \Delta h$ (Mild 0.2, Moderate 0.5, Severe 0.8). Visualization methods include before-and-after comparison charts, trajectory charts, heatmaps, risk-reward shift charts, and tornado diagrams. Robustness testing identifies resilient strategies via scenario grids (timing × severity × recovery rate).

Adoption Status: We adopted the scenario analysis framework and visualization methods, but independently implemented the health impact formula, designed specific injury scenarios, and developed the "load management insertion + critical battle concentration" strategy description.

15. GitHub *Copilot* (January 2026 version, Gemini3.0)

Used for Python code autocompletion, including Monte Carlo simulations, entropy weight calculations, Haversine distance formulas, CVaR computations, and data visualization. All code underwent manual verification and optimization, with algorithmic logic independently designed by the team.

Summary Statement

The AI tools listed above were utilized as auxiliary technologies during paper development to enhance productivity, improve language quality, and explore methodological options. All substantive modeling decisions, mathematical formulas, data analysis, result interpretations, and strategic recommendations were independently developed by our team at .The core contributions of our work—including the inner-outer loop framework design, dual-track player valuation with collaborative adjustment, application of the market erosion gravity model, heuristic ADP implementation optimized for season length, and integration of CVaR risk management—represent the authors' original intellectual work.

Each AI-generated output underwent critical evaluation, verification against authoritative sources, and adaptation to our specific research context. Throughout the report, we clearly distinguish between adopted content and our original contributions.